Project data

Project	5062 - Norton Hill / Silver St Toucan
Program date	17-11-2020
Version	1
Programmer	[redacted]
Country	UK
City	Midsomer Norton
Street1	Norton Hill Primary School
Street2	Silver Street
Controller type	PTC1-TR2500A
Controller board	EC2 32 Mb RAM
Controller Serial number	R12
Report created at	11/19/2020 7:00 PM
Database filename (.cpf)	5062_Norton_Hill_School_cfg_4041.cpf
Configurator version	12.0.2.0

Configuration Notes

Configuration Notes

```
Site - 5062 - Norton Hill / Silver St Toucan

Configuration number - 4041

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STRATOS Information:

Junction SCN                          - J99911
OTU SCN                               - X99910
[redacted]


[redacted]

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Special Notes:

* Ped demand 2 hours before tactile inactive DFM time to prevent spurious DFM's from non-use.
```

FACILITIES MODES AND PRIORITIES

FACILITIES

Facility	Enabled
Manual Control	Yes
Manual Step On Mode	No
CLF	Yes
UTC Facility	Yes
Hurry Call Mode	No
Priority	No
MOVA via UTC TO bits	Yes
MOVA M-inputs / PSVP	No

Hurry call (high priority) options

Use hurry call (high priority) mode for all red moves:	Yes
Part Time shutdown HC priority movements required:	No

MODES AND PRIORITY

Mode	PRIO	Dem. set leave	Dem. set enter
Hurry call (high priority)	1	-	-
Urban Traffic Control (UTC)	5	-	-
Hurry call (std priority)	-	-	-
Manual control	2	Start-up demand set	-
Cableless linking facility (CLF)	4	-	-
Vehicle actuated (VA)	6	-	-
Simple fix time (FT)	7	-	-
Public service vehicle priority	-	-	-
Selected cableless linking	3	Start-up demand set	Start-up demand set
Selected vehicle actuated	3	Start-up demand set	Start-up demand set
Selected fix time	3	Start-up demand set	Start-up demand set
ImFlow control	-	-	-
	-	-	-

Revertive Demand Sets

Phase	Type	RDC	Start-up	2	3	4	5	6	7	8
A	802 T: vehicle	A	Yes	No	No	No	No	No	No	No
B	802 T: vehicle	B	Yes	No	No	No	No	No	No	No
C	812 TN: toucan near side	-	Yes	No	No	No	No	No	No	No
D	815 W: warden box	-	Yes	No	No	No	No	No	No	No

STREAMS AND STAGES

STREAM

ID	Name	Type
1	STREAM1	Junction

STAGE

ID	Description	Stream No.	Demands AFTER Min Exp.	Demands BEFORE Min Exp.	Ripple Change	Startup stage	Arterial reversion stage	Switch off stage	All Red stage
1	STAGE1	1	No	Yes	No	Yes	Yes	Yes	No
2	STAGE2	1	No	Yes	No	No	No	No	No

Phases in stages

	A	B	C	D
1	X	X	-	-
2	-	-	X	X

PHASES

Types

Phase	Site Phase	Description	Type	Associated Phase
A	A	Silver Street N/B	802 T: vehicle	-
B	B	Silver Street S/B	802 T: vehicle	-
C	C	Pedestrians	812 TN: toucan near side	-
D	D	Bleepers	815 W: warden box	-

CONDITIONS

Phase	Tactile Interlock	Appearance type	Termination type
A	No	Always	At end of stage
B	No	Always	At end of stage
C	Yes	Always	At end of stage
D	No	Always	At end of stage

TIMINGS

Phase	Type	Min green	Min red	Start Amber	Amber	Ped Period V	Ped Period VI	Ped Period VII	Ped Period VIII	Pre-time max
A	802 T: vehicle	7	1	2	3					No
B	802 T: vehicle	7	1	2	3					No
C	812 TN: toucan near side	5	1			3	10			No
D	815 W: warden box	3	1							No

Note: Use of zero second blackout Ped Perod 5 on Type 814 PD: is not current DfT policy and should be discouraged

PHASE GREEN TIMING RANGES

PHASE	MIN Lower Limit	MIN Upper Limit	MAX Lower Limit	MAX Upper Limit
A	7	15	0	120
B	7	15	0	120
C	4	9	0	0
D	3	9	0	0

PHASE TIMING SETS

Regular maximums

	1	2	3	4
A	30	20	30	20
B	30	20	30	20
C				
D				

Alternative maximums

	1	2	3	4
A				
B				
C				
D				

Variable blackout/red periods

	1	2	3	4
A				
B				
C				
D				

Minimum green

	1	2	3	4
A				
B				
C				
D				

PSVP inhibition times

	1	2	3	4
A				
B				
C				
D				

PSVP maximum green times

	1	2	3	4
A				
B				
C				
D				

PHASE MATRICES

Settings

Starting intergreen	Handset maximum	Flashing Amber	All Red
9	30	0	0

Handset Int Offset	Default RLM Int	
0	2	

Opposing and conflicting

	A	B	C	D
A	–	–	C	C
B	–	–	C	C
C	C	C	–	–
D	C	C	–	–

Intergreen times

	A	B	C	D
A			6	0
B			6	0
C	5	5		
D	0	0		

Handset intergreen limits

	A	B	C	D
A			3	0
B			3	0
C	5	5		
D	0	0		

RLM additional intergreens

	A	B	C	D
A			2	
B			2	
C				
D				

RLM phase inhibits

	A	B	C	D
A	–	–	–	–
B	–	–	–	–
C	–	–	–	–
D	–	–	–	–

LAMP MONITORING

Applied sensing technology

Individual Monitoring Channels Used for RLUs ?	No

Lamp Switches

Phase	Type	SWR	SWA	SWG	SWWL
A	802 T: vehicle	R01	A01	G01	
B	802 T: vehicle	R02	A02	G02	
C	812 TN: toucan near side	R03		G03	A03
D	815 W: warden box			G04	

Phase Lamp Types

Phase	Description	Type	Red	Amber	Green	Wait
A	Silver Street N/B	T	T-LED 48 (D)	T-LED 48	T-LED 48	-
B	Silver Street S/B	T	T-LED 48 (D)	T-LED 48	T-LED 48	-
C	Pedestrians	TN	AGDMANCYCLE	-	AGDMANCYCLE	AGD_Wait
D	Bleepers	W	-	-	48 TEST	-

Lamp Monitor Settings

Phase	Description	Red 1	Red 2	Amber	Green
A	Silver Street N/B	Safety 1/2	None	Maintenance	Maintenance
B	Silver Street S/B	Safety 1/2	None	Maintenance	Maintenance
C	Pedestrians	Maintenance	None	None	Maintenance
D	Bleepers	None	None	None	None

Safety Lamp Monitor Shutdown Action

Phase	Description	Red 1	Red 2	Amber
A	Silver Street N/B	Shutdown	None	None
B	Silver Street S/B	Shutdown	None	None
C	Pedestrians	None	None	None

Phase	Description	Red 1	Red 2	Amber
D	Bleepers	None	None	None

PHASE DELAYS

FIXED TIME

FIXED TIME TO CURRENT MAXIMUM

STREAM	Fixed
STREAM1	Yes

Phase	Demand	Extend
A	Yes	Yes
B	Yes	Yes
C	No	No
D	No	No

STAGE MOVES

Move sets

Mode	SET
Hurry call (high priority)	1
Urban Traffic Control (UTC)	1
Hurry call (std priority)	0
Manual control	1
Cableless linking facility (CLF)	1
Vehicle actuated (VA)	1
Simple fix time (FT)	1
Public service vehicle priority	0

Set 1

	1	2
1	-	A
2	A	-

Set 2

	1	2
1	-	-
2	-	-

Set 3

	1	2
1	-	-
2	-	-

Set 4

	1	2

	1	2
1	–	–
2	–	–

Set 5

	1	2
1	–	–
2	–	–

Set 6

	1	2
1	–	–
2	–	–

Set 7

	1	2
1	–	–
2	–	–

Set 8

	1	2
1	–	–
2	–	–

MANUAL STAGE SELECTION

	1
But 1	1
But 2	2
But 3	-
But 4	-
But 5	-
But 6	-
But 7	-
But 8	-
But 9	-
But 10	-
But 11	-
But 12	-
But 13	-
But 14	-
But 15	-
But 16	-

Detectors

Application

ID	Detector	Type	Phase	Call	Cancel	Extend	Associated Det	DEM	CANCEL
1	AIN1	Queue detector	-				-	No	No
2	AX2	Vehicle loop	A			3.6	-	No	No
3	BIN3	Queue detector	-				-	No	No
4	BX4	Vehicle loop	B			3.8	-	No	No
5	CPBP1	Push button (kerbside A)	C				CKSP1	Yes	Yes
6	CPBP2	Push button (kerbside A)	C				CKSP1	Yes	Yes
7	CPBP3	Push button (kerbside A)	C				CKSP3	Yes	Yes
8	CPBP4	Push button (kerbside A)	C				CKSP3	Yes	Yes
9	CKSP1	Pedestrian kerbside detector	-		1		-	No	No
10	CKSP3	Pedestrian kerbside detector	-		1		-	No	No
11	CONCP2	Pedestrian on-crossing detector	C			1	-	No	No
12	CONCP4	Pedestrian on-crossing detector	C			1	-	No	No
13	CTACP1	IO1616 Detector	-				-	No	No
14	CTACP2	IO1616 Detector	-				-	No	No
15	CTACP3	IO1616 Detector	-				-	No	No
16	CTACP4	IO1616 Detector	-				-	No	No

Detector Fault Monitoring

ID	Detector	DFM Active Set 1	DFM Inactive Set 2	DFM Active Set 2	DFM Inactive Set 2	DFM Active Set 3	DFM Inactive Set 3	DFM Active Set 4	DFM Inactive Set 4	Detector DFM Error State	Detector Ok Count
1	AIN1	00:30	18:00	00:00	00:00	00:00	00:00	00:00	00:00	-	32
2	AX2	00:30	18:00	00:00	00:00	00:00	00:00	00:00	00:00	Occupied	32
3	BIN3	00:30	18:00	00:00	00:00	00:00	00:00	00:00	00:00	-	32
4	BX4	00:30	18:00	00:00	00:00	00:00	00:00	00:00	00:00	Occupied	32
5	CPBP1	00:10	254:00	00:00	00:00	00:00	00:00	00:00	00:00	-	4
6	CPBP2	00:10	254:00	00:00	00:00	00:00	00:00	00:00	00:00	-	4

ID	Detector	DFM Active Set 1	DFM Inactive Set 2	DFM Active Set 2	DFM Inactive Set 2	DFM Active Set 3	DFM Inactive Set 3	DFM Active Set 4	DFM Inactive Set 4	Detector DFM Error State	Detector Ok Count
7	CPBP3	00:10	254:00	00:00	00:00	00:00	00:00	00:00	00:00	-	4
8	CPBP4	00:10	254:00	00:00	00:00	00:00	00:00	00:00	00:00	-	4
9	CKSP1	01:00	18:00	00:00	00:00	00:00	00:00	00:00	00:00	Occupied	8
10	CKSP3	01:00	18:00	00:00	00:00	00:00	00:00	00:00	00:00	Occupied	8
11	CONCP2	00:30	18:00	00:00	00:00	00:00	00:00	00:00	00:00	Occupied	16
12	CONCP4	00:30	18:00	00:00	00:00	00:00	00:00	00:00	00:00	Occupied	16
13	CTACP1	00:05	254:00	00:00	00:00	00:00	00:00	00:00	00:00	-	4
14	CTACP2	00:05	254:00	00:00	00:00	00:00	00:00	00:00	00:00	-	4
15	CTACP3	00:05	254:00	00:00	00:00	00:00	00:00	00:00	00:00	-	4
16	CTACP4	00:05	254:00	00:00	00:00	00:00	00:00	00:00	00:00	-	4

IMFLOW

Enable ImFlow	Log to TDC
No	No

Settings

Wait time on deadlock (s)	Retries on UTC deadlock	Retries on UTC conflict
300	5	1

MASTER TIME CLOCK

MTC table

ID	Function	Arg1	Arg2	Start	End	Mo	Tu	We	Th	Fr	Sa	Su	G1	G2	G3	G4	G5	G6	G7	G8	Description
1	PTC1 Application	1	0	00:00:00	24:00:00	X	X	X	X	X	X	X	-	-	-	-	-	-	-	-	DO NOT ADJUST
2	Timing Set	4	0	00:00:00	24:00:00	X	X	X	X	X	X	X	-	-	-	-	-	-	-	-	Midnight Timing Set
3	Timing Set	1	0	07:30:00	24:00:00	X	X	X	X	X	X	X	-	-	-	-	-	-	-	-	
4	Timing Set	2	0	09:30:00	24:00:00	X	X	X	X	X	X	X	-	-	-	-	-	-	-	-	
5	Timing Set	3	0	15:30:00	24:00:00	X	X	X	X	X	X	X	-	-	-	-	-	-	-	-	
6	Timing Set	2	0	18:30:00	24:00:00	X	X	X	X	X	X	X	-	-	-	-	-	-	-	-	
7	Timing Set	4	0	20:00:00	24:00:00	X	X	X	X	X	X	X	-	-	-	-	-	-	-	-	
8	Audible	1	0	07:30:00	22:30:00	X	X	X	X	X	X	X	-	-	-	-	-	-	-	-	Bleeper Timetable

CONFLICT EXTENSION RED

CABLELESS LINKING FACILITY

Global settings

Setting	Value
Sync. Mode	Daily
Sync. Day	Monday
Ref. time	06:00

Plans

Functions and actions

OTU

Units

General Integral OTU options

Discrete OTU

ID	Invert control bits	Invert reply bits
1	No	No

Control and reply bits

Control / Reply

Function	Arg	Label	Position ID	Label	Arg	Function
F	1	F1	1	G1	1	G
FD	2	FD2	2	G2	2	G
D	2	D2	3	SD2	2	SD
DX	1	DX1	4	DF	0	DF
TS	0	TS	5	CS	0	TSR
-	0		6	RR	1	RR
-	0		7	CF	0	CF
-	0		8	LF1	1	LF
-	0		9	LF2	2	LF
-	0		10	LE	0	LE
-	0		11		0	-
-	0		12		0	-
-	0		13		0	-
-	0		14		0	-
-	0		15		0	-
SF	1	TO1	16	CRB1	1	UF

UTC Demand/Extend bits - Phases/Stages allocation

Bit	Phases Demanded	Phases Extended	Stages Demanded	Stages Extended

Bit	Phases Demanded	Phases Extended	Stages Demanded	Stages Extended
D 2	C			
DX 1	A,B,C			

Keep default Demand reply options

Default Stage Demand (SD) reply	No
Default Phase Demand (DR) reply	No

RTC

Synchronisation time (UTC TS input)	02:00
Confirm time (UTC RT output)	0:0:0

Special UTC Reply bits

	G1/G2	RR
Manual mode operative	No	Yes
Manual mode selected	No	Yes
No lamp power (excluding RLM and PT)	No	n/a
Normal not selected on the manual panel	No	Yes

CG reply bit weekday-coded	No

UG405 bit mapping

Settings

Enable UG405	No
OTU System Code Number(SCN)	

Control / Reply

ID	Label	Controller SCN	UG405 Name	Bit Index	Label	Controller SCN	UG405 name	Bit index
1	F1		-		G1		-	
2	FD2		-		G2		-	
3	D2		-		SD2		-	

ID	Label	Controller SCN	UG405 Name	Bit Index	Label	Controller SCN	UG405 name	Bit index
4	DX1		-		DF		-	
5	TS		-		CS		-	
6			-		RR		-	
7			-		CF		-	
8			-		LF1		-	
9			-		LF2		-	
10			-		LE		-	
11			-				-	
12			-				-	
13			-				-	
14			-				-	
15			-				-	
16	TO1		-		CRB1		-	

HURRY CALLS

PUBLIC SERVICE VEHICLE PRIORITY

SPEED ASSESSMENT AND SPEED DISCRIMINATION

APPLICATION BUILDING BLOCKS

Software Switches (User Parameters)

ID	Name	Default	Min Value	Max Value
1	Enable_PTM	0	0	1

Event Pulses

ID	Name	Type	Input Type	On	OFF
1	CRB1	Wave	Level	2	180

Event Pulse Input Conditioning
evp1() = (macm(0)>1) && (stgc(0)!=0) && mpauto(0) && (mUTC(0)==0) && (mPSVP(0)==0);

Event Filter Input Conditioning

SPECIAL CONDITIONING - (VM Functions)

O.T.U. Control & Reply Bit Special Conditioning

```
rf_2(arg) = cfa;

rf_7(arg)
if (arg == 2) then return (dr(C)); endif
return FALSE;
end

rf_16(arg)
if (arg==1) then return lf (-1,1911); endif
if (arg==2) then return lf (-1,7); endif
end

rf_27(arg)
if (arg == 2) then return (dr(C)); endif
return FALSE;
end

rf_28(arg)
if (arg == 1) then return (dr(A) || dr(B) || dr(C)); endif
return FALSE;
end

rf_32(arg)
if (arg==1) then
  return ((macm(0)>4) && ((mpauto(0) && (((mPSVP(0) || evp(CRB1)) && (mUTC(0)==0)) || (in(utciTO1) && ufac(0)))) == 1));
endif
end

rf_34(arg) = (macm(xp) != 6);

rf_40(arg) = mPSVP(xp);

urG1() = (stgc(0)==1) && (stgr(0,0,0));

urDR1() = dr(A) || dr(B);

urG2() = (stgc(0)==2) && (stgr(0,0,0));

urDR2() = dr(C) || dr(D);
```

Integral O.T.U. Special Conditioning
`otu_dstate(d) = get(h_xdet_sts, d) & DET_BEZET_MASK;` `otu_dfault(d) = get(h_xdet_sts, d) & DET_FAULT_MASK;` `otu_dcnt(d) = get(h_xdet_cnt, d);`

U.T.C. (G1/G2) Special Conditioning

CLF Request & Inhibit Special Conditioning

P.S.V.P. Pre Check-in, Check-in & Check-out Special Conditioning

Hurry Call Delay, Force, Demand & Inhibit Special Conditioning

Phase Delay Appearance Special Conditioning

Stream On/Off Control Special Conditioning

Stream On/Off Control Special Conditioning

```
roffsync() = okoff(0);

mac1_swon(t) = yellow_period(0,t);

macon1() = (minon(xp)==0);

macoff1() = ((macm(xp) <= 1) && (mact(xp) <= tson(7)));
```

Detector Count Activity Window Special Conditioning

```
nokbs()

var det;
for det=0 to (nrel(h_dfunc) -1) do
  if ddo(det) && (fcteg(get(h_dsg,det))>0) && (ddr(det)!=1) && (get(h_dfunc,det)==25) then
    put(h_dfunc,det,65);
  endif
  if (get(h_dfunc,det)==65) && (((ddo(det) || (dr(get(h_dsg,det))))!=1) || fcg(get(h_dsg,det))) then
    put(h_dfunc,det,25);
  endif
endfor
end

dact_CONCP2() = fcg(C) || fcr(C) || (fcbo(C) >= 3);

dact_CONCP4() = fcg(C) || fcr(C) || (fcbo(C) >= 3);
```

Phase Control Special Conditioning

Phase Control Special Conditioning

```
latch(ph) = dx(ph) && (fcg(ph)==0);

pd_ALL(ph)

if ((xsf(XSF_PSET_ERR) != AUTOSET_STATE_DONE) && (xsf(XSF_PSET_ERR) != AUTOSET_STATE_ERROR)) then return (1); endif
end

pe_ALL(ph) = ngp1(ph);

pd_A() = TRUE;

pd_B() = TRUE;

wl_C() = dr(C) || fci(C);

pd_C() = uD(2) || uDX(0) || pd_ALL(C) || TacDem(C,CTACP1) || TacDem(C,CTACP2) || TacDem(C,CTACP3) || TacDem(C,CTACP4) || latch(C);

pvbo_C() = ((dact(CONCP2)==0) || dde(CONCP2) || (ddg1(CONCP2)==0)) || ((dact(CONCP4)==0) || dde(CONCP4) || (ddg1(CONCP4)==0));

taco_C(a) = cout(AUDIO,a);

pd_D() = FALSE;

pa_D() = (sgs(C)==3) && out(AUDIO);
```

Phase Timing Set Selection Special Conditioning

Dummy Detector Special Conditioning

Mode Control Special Conditioning

Mode Control Special Conditioning

```
rHCH() = rar(xp);

rUTC()
if (xp==0) then
  if in(utciTO1) then return ufac(xp); endif
endif
return ufav(xp)||ufpv(xp);
end

rMAN() = mpman(xp);

rCLF()

if (xp==0) then
  return clfp && clfv(xp,clfp) && (clfi == 0);
endif
end

rVA() = 1;

rFT() = 1;

rSCLF()

if (xp==0) then
  return clfp && clfv(xp,clfp) && (mpclf(xp) || clfmp);
endif
end

rSVA() = mpva(xp);

rSFT() = mpft(xp);

tlcfa_ImFlow_overrule(a) = overruleIMFLOW(a);     /* Check whether ImFlow is currently overruled */
```

All Red Detection Operation Special Conditioning

Manual Panel Stage LED Conditions

```
MpStageLEDsNoDefault()
mpdiso(20,1);
```

Manual Panel Stage LED Conditions

```
mpdiso(21,1);
mpdiso(22,1);
mpdiso(23,1);
mpdiso(24,1);
mpdiso(25,1);
mpdiso(26,1);
mpdiso(27,1);
mpdiso(28,1);
mpdiso(29,1);
mpdiso(30,1);
mpdiso(31,1);
mpdiso(32,1);
mpdiso(33,1);
mpdiso(34,1);
mpdiso(35,1);
end

DriveMpStageLEDs()
MpStageLEDsNoDefault();
mpledfunc20();
mpledfunc21();
mpledfunc22();
mpledfunc23();
mpledfunc24();
mpledfunc25();
mpledfunc26();
mpledfunc27();
mpledfunc28();
mpledfunc29();
mpledfunc30();
mpledfunc31();
mpledfunc32();
mpledfunc33();
mpledfunc34();
mpledfunc35();
end

mpledfunc20() = cmled(20,(stgc(0)==1)*(2-stgr(0,1,0)));

mpledfunc21() = cmled(21,(stgc(0)==2)*(2-stgr(0,1,0)));

mpledfunc22() = 0;

mpledfunc23() = 0;

mpledfunc24() = 0;

mpledfunc25() = 0;
```

Manual Panel Stage LED Conditions

```
mpledfunc26() = 0;

mpledfunc27() = 0;

mpledfunc28() = 0;

mpledfunc29() = 0;

mpledfunc30() = 0;

mpledfunc31() = 0;

mpledfunc32() = 0;

mpledfunc33() = 0;

mpledfunc34() = 0;

mpledfunc35() = 0;
```

Initialisation & General Special Conditioning

```
init()
open (dfmoff);
open_handles();
return 1;
end

dfm_f() = sf1(-1) || fci(-1);
```

User Defined VM Functions

User Defined VM Functions

```
post_100ms()
dfm_led;
nokbs;
mova_dets;
LEDs;
Bleeper_Output;
NOT_PTM;

end

dfm_led()
var x;
for x=0 to (nrel(h_dfunc) -1) do
cmled(36,(dde(x) && (rf(-1)!=1)) || (rf(-1) && (tsec() & 1)));
endfor
end

mova_dets()
cout(MDET1,ddo(AIN1));
cout(MDET2,ddo(AX2));
cout(MDET3,ddo(BIN3));
cout(MDET4,ddo(BX4));
cout(MDET5,dr(C) && (fci(C)==0));

end

LEDs()
cmled(7,out(AUDIO) + out(Bleeper_Switch));
cmled(14,(mUTC(0)) + (2*(mUTC(0) && in(utciTO1))));
end

Bleeper_Output() = cout(Bleeper_Switch,(out(AUDIO) && (sgs(C)==3)));

NOT_PTM()
rmx(A,(dr(C)==0) && (UP(Enable_PTM)==0) && (mFT(0)==0));
rmx(B,(dr(C)==0) && (UP(Enable_PTM)==0) && (mFT(0)==0));
end
```

Stage Moves

Hurry Call High Mode - HCH (Normally for Part Time / Prom Swap Facility)

Hurry Call Standard Mode - HCL

Vehicle Actuated Mode - VA

```
mVA1_2d() = dSTGd(1, 2);

mVA1_2e() = dSTGe(1, 2);

mVA1_2i() = FALSE;

mVA2_1d() = dSTGd(2, 1);

mVA2_1e() = dSTGe(2, 1);

mVA2_1i() = FALSE;

mVA2_1du4() = dn();

mVA2_1eu4() = dSTGe(2, 1);

mVA2_1iu4() = FALSE;
```

Fixed Time Mode - FT

Fixed Time Mode - FT

```
mFT1_2d() = dSTGd(1, 2);

mFT1_2e() = dSTGe(1, 2);

mFT1_2i() = FALSE;

mFT2_1d() = dSTGd(2, 1);

mFT2_1e() = dSTGe(2, 1);

mFT2_1i() = FALSE;

mFT2_1du4() = dn();

mFT2_1eu4() = dSTGe(2, 1);

mFT2_1iu4() = FALSE;
```

Cableless Linking Mode - CLF

```
mCLF1_2d() = dCLFd(1, 2);

mCLF1_2i() = dCLFi(1, 2);

mCLF2_1d() = dCLFd(2, 1);

mCLF2_1i() = dCLFi(2, 1);
```

Urban Traffic Control Mode - UTC

```
mUTC1_2d() = dUTCd(1, 2);

mUTC1_2i() = dUTCi(1, 2);

mUTC2_1d() = dUTCd(2, 1);

mUTC2_1i() = dUTCi(2, 1);
```

Public Service Priority Mode - PSVP

Manual Control Mode - MAN

```
mMAN1_2d() = mstg(xp)==2;

mMAN2_1d() = mstg(xp)==1;
```

General (default) Stage Move Conditions

```
dSTGd(f,t) = drs(t);

dSTGe(f,t) = exm(t);

dUTCd(f,t)
if t==1 then return uF(1) || (uFD(1) && (uD(1) || in(utciTO1))); endif
if t==2 then return uF(2) || (uFD(2) && ((uD(2) || dr(C)) || in(utciTO1))); endif
return uF(t) || (uFD(t) && (uD(t) || drs(t)));
end

dUTCi(f,t) = ((uF(f) || uFD(f)) && ((uGO(xp)==0) || ex(t)));

dCLFd(f,t) = cIM(t) || (cDD(t) && drs(t)) || (cPS(t) && drs(t));

dCLFi(f,t) = cIM(f) || cDD(f) || cH(xp) || (cPS(t) && ex(t));

dPSVPd(f,t) = drs(t);

dPSVPe(f,t) = exm(t);

dSTGd1() = dr(A) || dr(B);

dSTGe1() = er(A) || er(B);

dPSVPd1() = pdp(A) || pdp(B);

dPSVPe1() = pep(A) || pep(B);

dUTCd1(t) = uF(t) || (uFD(t) && (uD(t) || dr(A) || dr(B)));

dUTCi1(f,t) = ((uF(f) || uFD(f)) && ((uGO(xp)==0) || ex(t)));
```

General (default) Stage Move Conditions

```
dCLFd1(t) = cIM(t) || (cDD(t) && (dr(A) || dr(B))) || (cPS(t) && (dr(A) || dr(B)));

dCLFi1(f,t) = cIM(f) || cDD(f) || cH(xp) || (cPS(t) && ex(t));

dSTGd2() = dr(C) || dr(D);

dSTGe2() = er(C) || er(D);

dPSVPd2() = pdp(C) || pdp(D);

dPSVPe2() = pep(C) || pep(D);

dUTCd2(t) = uF(t) || (uFD(t) && (uD(t) || dr(C) || dr(D)));

dUTCi2(f,t) = ((uF(f) || uFD(f)) && ((uGO(xp)==0) || ex(t)));

dCLFd2(t) = cIM(t) || (cDD(t) && (dr(C) || dr(D))) || (cPS(t) && (dr(C) || dr(D)));

dCLFi2(f,t) = cIM(f) || cDD(f) || cH(xp) || (cPS(t) && ex(t));
```

SYSTEM PARAMETERS

UK Parameters

Name	Description	Min	Max	Def	Value
MAN_TIMEOUT	Manual control timeout	60	600	600	600
MAN_DEMAND_ERROR	Duration of the manual demand error indication	0	60	5	5
MAN_ENABLE	Manual control enabled.	0	1	1	1
DFM_FUNC	Bit mask specifying behaviour of DFM indicator	0	999	1	1
CLF_TIMER_SYNC	Duration of the CLF group Timer Synchronisation Signal	0	0	0	0
UTC_TS	Time in HH:MM used for COTU_TS	0	2359	1200	200
UTC_FORCE_TIMEOUT	The force bit watchdog timeout	120	300	200	200
UTC_FORCE_ACCEPT	The force accept timeout in system ticks.	1	10	4	4
UTC_LO_DELAY	Delay in seconds before COTU_LO changes is accepted.	0	99	10	10
UTC_RT_HOUR	Hours used for RT reply bit (-1 = any hour).	-1	23	0	0
UTC_RT_MIN	Minutes used for RT reply bit (-1 = any minute).	-1	59	0	0
UTC_RT_SEC	Seconds used for RT reply bit	-1	59	0	0
UTC_RT_DURATION	Duration of the RT reply bit in seconds	1	10	3	3
UTC_CG_DURATION	Duration of the CG reply bit in seconds	1	10	3	3
UTC_TSR_DURATION	Duration of the TSR reply bit in seconds	1	10	3	3
UTC_G1G2_FUNC	Bit mask specifying behaviour of G1/G2 reply bits	0	255	1	0
UTC_RR_FUNC	Bit mask specifying behaviour of RR reply bit	0	255	14	14
UTC_PV_ACCEPT	The PV accept time in system ticks	0	255	0	0
UTC_PV_HOLD	The PV hold time in system ticks	0	255	40	40
UKSTG_DIM_ALARM	The maximum time [in hours] for dimmed operation (-1 = solar cell disabled, 0 = enabled without timeout processing)	-1	100	24	24
UKSTG_DIM_FILTER	The call/cancel delay [in sec] for the DIM relays	0	255	15	15
UTC_CRB_PULSE_ON_TIME	CRB On Timer	0	1800	600	600
UTC_CRB_PULSE_OFF_TIME	CRB Off Timer	0	10	2	2

System Parameters

Name	Description	Min	Max	Value

Name	Description	Min	Max	Value
MMI8408	XMMI: 8 x 40 MMI	0	1	1
XIN_L	XIN_L: Number of IN logging lines	1	999	10
XOUT_L	XOUT_L: Number of OUT logging lines	1	999	10
XDET_L	XDET_L: Number of DET logging lines	1	999	100
XDET_F	XDET_F: Status in case of fault	0	2	0
XSG_L	XSG_L: Number of SG logging lines	0	999	100
XHTTP	XHTTP: Presence of web server	0	1	1
XHTTP_PORT	XHTTP_PORT: Webserver port	0	65535	80
XLM_MAL	XLM_MAL: Go to major alarm if power reference(s) are not set	0	1	0
XLM_RLM	XLM_RLM: RLM callback interval in 0.1 [s]	0	100	10
XLM_T	XLM_T: Tracking off / on	0	20	5
XLM_TF	XLM_TF: Tracking filter	1	20	8
XLM_NV	XLM_NV: Nominal (bright) voltage	0	240	230
XLM_MON	XLM_MON: Monitoring type	0	10	3
XLM_AC	XLM_AC: Automatic calibration	0	1	1
XLM_DIM	XLM_DIM: Bright Din Calibration off / on	0	1	1
CLF_SYNC	CLF Synchronisation option.	0	3	0
CLF_SYNC_WDAY	CLF Weekly sync is done on..	1	7	1
CLF_TIMER_SYNC	Duration of the CLF group Timer Synchronisation Signal	0	2400	600
CLF_NON_BASE_TIME	Select base time or non base time (0 or 1)	0	1	1
XIOTU_RTS	RTS activated after # received characters (use 1, 2 or 3)	1	3	1
XIOTU_SCOOT	Default bit number used for scoot counted detectors (0 to 7)	0	7	0
XIOTU_CTS	CTS timeout per 10ms	4	10	6
UKMP_TYPE	UK Manual panel type (0=std, 1=ped, 2=multiple stages)	0	2	1
ENGTERM_BAUD	Engterm baudrate	0	99999	9600
HCH_ALLRED	Use HCH for all red stage moves	0	1	1
HCH_SWOFF	Use HCH for switch off stage moves	0	1	0
PT_SYNC	Synchronize move to part time mode	0	1	1
CLF_MANUAL_STEP	Enable the Manual Step	0	1	0
CLF_DEFAULT_PLAN	None	-1	999	0
CLF_MANUAL_MEMOS	Memo for the manual step	0	1	0
COMPRESS_HARDWARE	Compress Hardware	0	1	0
XWDS_SENSOR_TIMEOUT	Wireless Detection sensor time out value	0	65535	180

Name	Description	Min	Max	Value
ALERTPLUS_ENABLE	Alert+: Enable the service	0	1	0
ALERTPLUS_NRSESSIONS	Alert+: Number of sessions	1	99	2
ALERTPLUS_LOGINUSERG1	Alert+: Login code for user group 1	0	9999	0
ALERTPLUS_LOGINUSERG2	Alert+: Login code for user group 2	0	9999	0
ALERTPLUS_LOGINUSERG3	Alert+: Login code for user group 3	0	9999	0
ALERTPLUS_LOGINUSERG4	Alert+: Login code for user group 4	0	9999	0
IMFLOW_ENABLE	ImFlow enabled	0	1	0
IMFLOW_WDG_TIME	ImFlow: Wait time for deadlock supervision [s]	1	999	300
IMFLOW_MAX_WDG_RETRIES	ImFlow: Max. retries on deadlocks in UTC control	1	99	5
IMFLOW_MAX_CONFL_RETRIES	ImFlow: Max. retries on conflicts in UTC control	1	99	1
IMFLOW_SIM_IF	ImFlow Sim. interface	0	1	1
IMFLOW_TDC	ImFlow TDC logging	0	1	0
XWIFI	WiFi service activated [FlowNode]	0	1	0

Traffic Data Collector (TDC)

TDC enabled	Persistent	Readonly	MF (max. files)	Assigned memory
No	0	0	0	1 Mb

MEMO FIELDS

HISTORY	
Description	Text in the History text block between: --CREDAT-- and --UPDATE--
Contents	* This is memofield HISTORY

B_HISTORY	
Description	Text in the History text block between: --CREDAT-- and --UPDATE-- (AMSEC1.CNF)
Contents	; This is memofield HISTORY_B

ABC_INC	
Description	Code at the end of the file before the end of file remark (AMSEC1.CNF)
Contents	; This is memofield AMSEC_INC

XP1_INC	
Description	Code at the begin of the file, just below History, under the heading 'XP1_INC'.
Contents	**/* This is memofield XP1_INC** */

XP2_INC	
Description	Code under the heading 'XP2_INC', just before the 'XIN INPUTS' definitions.
Contents	**/* This is memofield XP2_INC** */

XP3_INC	
Description	Code under the heading 'XP3_INC', at the end of the file.

XP3_INC	
Contents	**/* This is memofield XP3_INC** */ P("MPDISO.R1") { P(0);D(1); P(1);D(1); P(2);D(1); P(3);D(1); P(4);D(1); P(5);D(1); P(6);D(1); P(7);D(1); P(14);D(1); P(26);D(1); }

SADAT_INC	
Description	Code at end of the file. (SADAT.CNF)
Contents	**/* This is memofield SADAT_INC** */ **/* Function Stops CER Dets Extending phase A** */ P("DFUNC.R30") { **/* P(Clearance Det); D(0b00000000);** */ }

VMFUNC_INC1	
Description	VMFUNC process conditions
Contents	**/* This is memofield VMFUNC_INC1** */ handle dfmoff = "XDETN.R11"; function TacDem(2);

VMFUNC_INC2	
Description	VMFUNC process conditions

VMFUNC_INC2	
Contents	(see code below)

```
/* This is memofield VMFUNC_INC2 */

/* Demand Ped Function If Not used with TAC Inactive DFM Time */

TacDem(phs,det)
if (sgs(phs)==1) && (dde(det)==0)  then
   if fct(phs)>(((get(dfmoff,det)/100)-2)*36000) then
      return TRUE;
   else
      return FALSE;
   endif
endif
end
```

HARDWARE CONFIGURATION

Device counts

Lamp control devices	Count	Detection devices	Count	I/O devices	Count
VIRTUAL-LCM	1	ED316	0	IO1616	3
LCM	1	MTS4E	1	RIO	0
RLU	0	WDS	0		
RLU-9	0	FLIR-ZONE	0		
		FLIR-OUTPUT	0	XSER6	0
XLS12	0	XD16	0	XIO16	0
Dummy-Phase	0	Dummy-Detector	0		

Optional devices	Setting
Manual Panel Type	Pedestrian
Manual Panel Flashing	Disabled
Solar Cell Monitor	24 hour time out
Dimming Operating	ELV 24V Solar Cell

Configuration - LCM / XLS12 / RLU

```
LCM
--- Physical LCM: 01 ---
 001: A/Red   (R01)
 002: A/Amber (A01)
 003: A/Green (G01)
 004: B/Red   (R02)
 005: B/Amber (A02)
 006: B/Green (G02)
 007: C/Red   (R03)
 008: C/Wait  (A03)
 009: C/Green (G03)
 012: D/Green (G04)
```

Configuration - MTS4E / ED316

```
MTS4E
--- Unit: 01 ---
```

```
001: Detector AIN1
002: Detector AX2
003: Detector BIN3
004: Detector BX4
```

IOT State - Lamp control

```
LCM

Unit    Addr  *
======= ====  ==============
 01      001  A/Red   (R01)
 01      002  A/Amber (A01)
 01      003  A/Green (G01)
 01      004  B/Red   (R02)
 01      005  B/Amber (A02)
 01      006  B/Green (G02)
 01      007  C/Red   (R03)
 01      008  C/Wait  (A03)
 01      009  C/Green (G03)
 01      012  D/Green (G04)
```

```
RLU

Unit    Addr  *
======= ====  ==============
```

IOT State - Detection

```
MTS4E

Unit    Addr  *
======= ====  ==============
 01      001  AIN1
 01      002  AX2
 01      003  BIN3
 01      004  BX4
```

IOT State - I/O

```
IO1616-IN

Unit    Addr  *
======= ====  ==============
 01      001  CPBP1
 01      002  CPBP2
 01      003  CPBP3
 01      004  CPBP4
```

```
IO1616-OUT

Unit    Addr  *
======= ====  ==============
 01      001  MDET1
 01      002  MDET2
 01      003  MDET3
 01      004  MDET4
```

```
01      005  CKSP1
01      006  CKSP3
01      007  CONCP2
01      008  CONCP4

02      001  CTACP1
02      002  CTACP2
02      003  CTACP3
02      004  CTACP4

03      001  UTC_I1
03      002  UTC_I2
03      003  UTC_I3
03      004  UTC_I4
03      005  UTC_I5
03      006  UTC_I6
03      007  UTC_I7
03      008  UTC_I8
03      009  UTC_I9
03      010  UTC_I10
03      011  UTC_I11
03      012  UTC_I12
03      013  UTC_I13
03      014  UTC_I14
03      015  UTC_I15
03      016  UTC_I16
```

```
02      001  MDET5

03      001  UTC_O1
03      002  UTC_O2
03      003  UTC_O3
03      004  UTC_O4
03      005  UTC_O5
03      006  UTC_O6
03      007  UTC_O7
03      008  UTC_O8
03      009  UTC_O9
03      010  UTC_O10
03      011  UTC_O11
03      012  UTC_O12
03      013  UTC_O13
03      014  UTC_O14
03      015  UTC_O15
03      016  UTC_O16
```

INPUTS and OUTPUTS

Detectors

ID	Detector Name	Invert	Unit Type	Unit Pos.
1	AIN1	No	MTS4E-1	1
2	AX2	No	MTS4E-1	2
3	BIN3	No	MTS4E-1	3
4	BX4	No	MTS4E-1	4
5	CPBP1	No	IO1616-1	1
6	CPBP2	No	IO1616-1	2
7	CPBP3	No	IO1616-1	3
8	CPBP4	No	IO1616-1	4
9	CKSP1	Yes	IO1616-1	5
10	CKSP3	Yes	IO1616-1	6
11	CONCP2	Yes	IO1616-1	7
12	CONCP4	Yes	IO1616-1	8
13	CTACP1	No	IO1616-2	1
14	CTACP2	No	IO1616-2	2
15	CTACP3	No	IO1616-2	3
16	CTACP4	No	IO1616-2	4

Inputs

ID	Input	Label	Invert	Toggle	Unit Type	Unit Pos.
1	UTC_I1	F1	No	No	IO1616-3	1
2	UTC_I2	F2	No	No	IO1616-3	2
3	UTC_I3	D1	No	No	IO1616-3	3
4	UTC_I4	DX1	No	No	IO1616-3	4
5	UTC_I5	TS	No	No	IO1616-3	5
6	UTC_I6		No	No	IO1616-3	6
7	UTC_I7		No	No	IO1616-3	7
8	UTC_I8		No	No	IO1616-3	8
9	UTC_I9		No	No	IO1616-3	9

ID	Input	Label	Invert	Toggle	Unit Type	Unit Pos.
10	UTC_I10		No	No	IO1616-3	10
11	UTC_I11		No	No	IO1616-3	11
12	UTC_I12		No	No	IO1616-3	12
13	UTC_I13		No	No	IO1616-3	13
14	UTC_I14		No	No	IO1616-3	14
15	UTC_I15		No	No	IO1616-3	15
16	UTC_I16	TO1	No	No	IO1616-3	16
17	DIM_IN		No	No	DIMMING	1

Outputs

ID	Output	Label	Invert	Unit Type	Unit Pos.	Intersection	SRM2 Only
1	MDET1		No	IO1616-1	1	0	No
2	MDET2		No	IO1616-1	2	0	No
3	MDET3		No	IO1616-1	3	0	No
4	MDET4		No	IO1616-1	4	0	No
5	MDET5		No	IO1616-2	1	0	No
6	Bleeper_Switch		No	-	0	0	No
7	UTC_O1	G1	No	IO1616-3	1	0	No
8	UTC_O2	G2	No	IO1616-3	2	0	No
9	UTC_O3	SD2	No	IO1616-3	3	0	No
10	UTC_O4	DF	No	IO1616-3	4	0	No
11	UTC_O5	TSR	No	IO1616-3	5	0	No
12	UTC_O6	RR	No	IO1616-3	6	0	No
13	UTC_O7	CF	No	IO1616-3	7	0	No
14	UTC_O8	LF1	No	IO1616-3	8	0	No
15	UTC_O9	LF2	No	IO1616-3	9	0	No
16	UTC_O10	LE	No	IO1616-3	10	0	No
17	UTC_O11		No	IO1616-3	11	0	No
18	UTC_O12		No	IO1616-3	12	0	No
19	UTC_O13		No	IO1616-3	13	0	No
20	UTC_O14		No	IO1616-3	14	0	No
21	UTC_O15		No	IO1616-3	15	0	No

ID	Output	Label	Invert	Unit Type	Unit Pos.	Intersection	SRM2 Only
22	UTC_O16	CRB1	No	IO1616-3	16	0	No
23	AUDIO		No	-	0	1	No
24	DIM_OUT		No	DIMMING	1	0	No

Report executed at 11/19/2020 7:00 PM